Agenda

- Intro and problem Statement
- What is Time-series and Forecasting
- Preprocess Time-Series Data
  - Handling missing Values
  - Dealing with Anomalies
- Breaking Down TS into different Component
  - Trend
  - seasonality
- Moving Average

Module Overview

6- Time-series

- Intro to TS
- 5 session of TS-Algos

Recommendation System

5 Sessions

Practical session : Project Based

Problem Statement

You are D.S at Mobiplus

Mobile Orders

Date	Sales
1-1-2001	203
1-2-2001	500
1-3-2001	605
1-4-2001	⋮
⋮	

Agenda 1: Is there a pattern in Data

Agenda 2: We need to forecast/predict the demand with

MAPE of not more than 5%

mean absolute percentage Error

Agenda 3: Need to supplement Point forecast with Range Forecast

Target Variable

Continous Variable

(Regression)

- Point (Precise) → Single Value
- Range → 11000-12000 95% Confidence

Predictor → Independent Vars

Target → Variable

	DATE	Sales
0	2001-01-01	6519.0
1	2001-02-01	6654.0
2	2001-03-01	7332.0
3	2001-04-01	7332.0
4	2001-05-01	8240.0

T.C.S @ Tmow morning ?.

Forecasting:

Features are previous Values of Target Variable.

Why sale forecast is important ?

- Manage Staff and Inventory
- Logistics
- Promotions and Deals
- Financial Strategy

Over Forecasting Side-Effect:

- Overstocking
- Inventory holding and maintenance

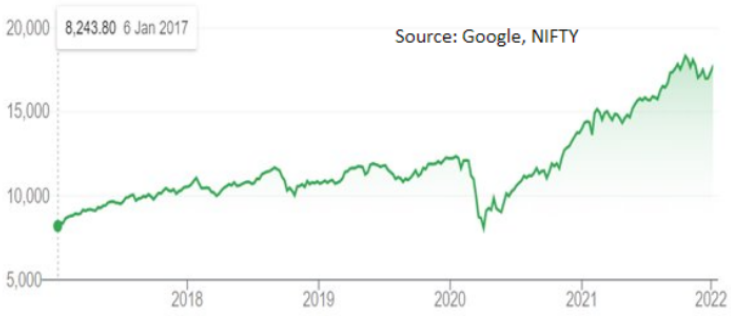
Under Forecasting Side-Effect:

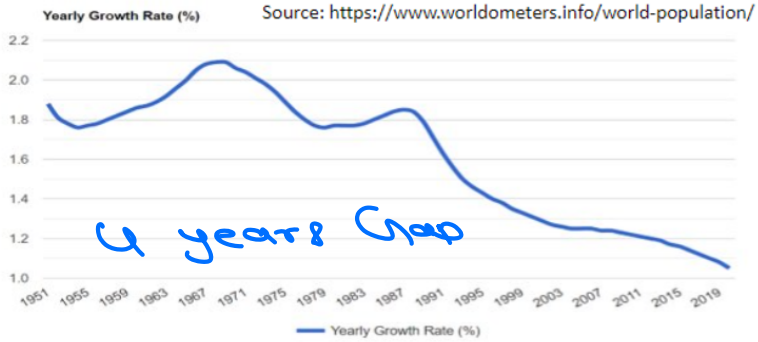
- Lot of Unmet Demands and Unsatisfied customer
- Loss of Brand image and Goodwill
- Loss of Revenue

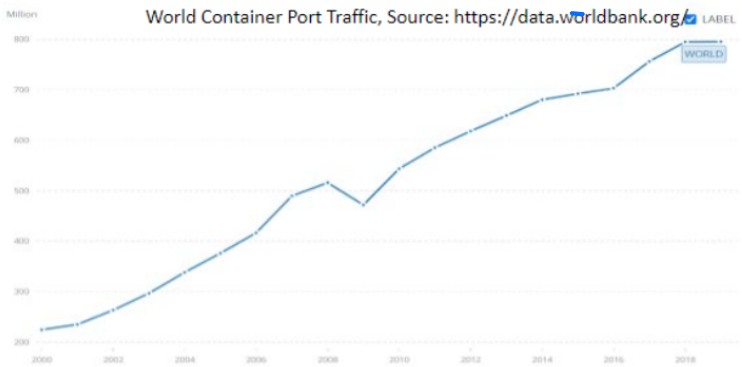
What is Time Series Data?

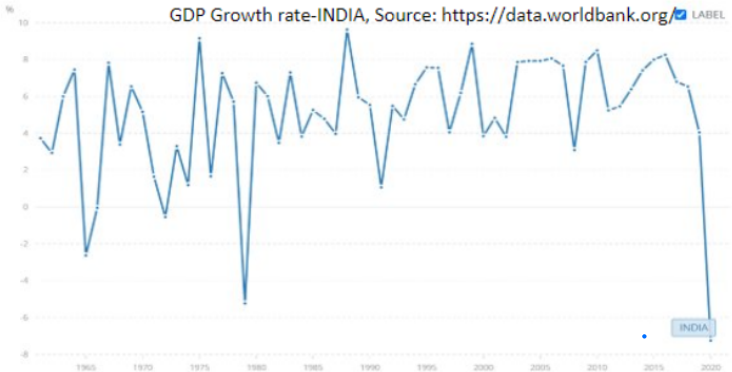
A signal indexed by Ordered Timestamp

- index (ordered TS)
- signal (Target)









Forecasting Model

Task: $y_{t+1} \leftarrow f(y_t, y_{t-1}, y_{t-2} \cdots)$

(Historical Data)

to 2023

$y_{2024} \leftarrow (y_{2023}, y_{2022}, \cdots)$

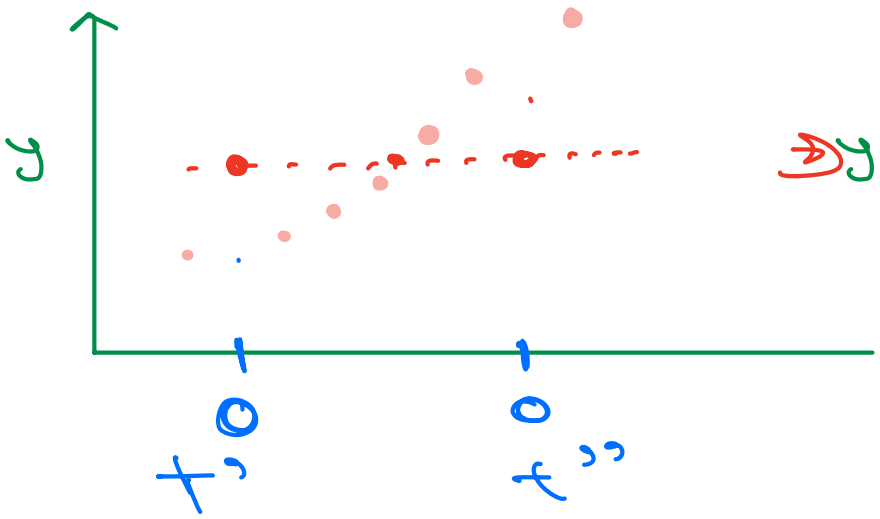
* Note: The Time-series column must be Uniformly seperated (No missing Dates should be there

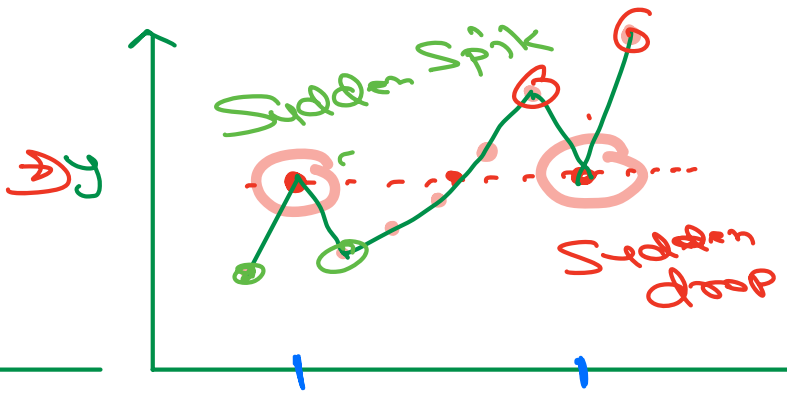
Issues with T.S Data

- Missing Sales Value
- Outliers

Treating Missing Values

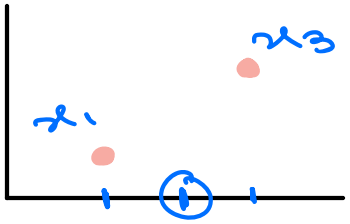
- Mean, Median, 0-fill (Static Replacement)



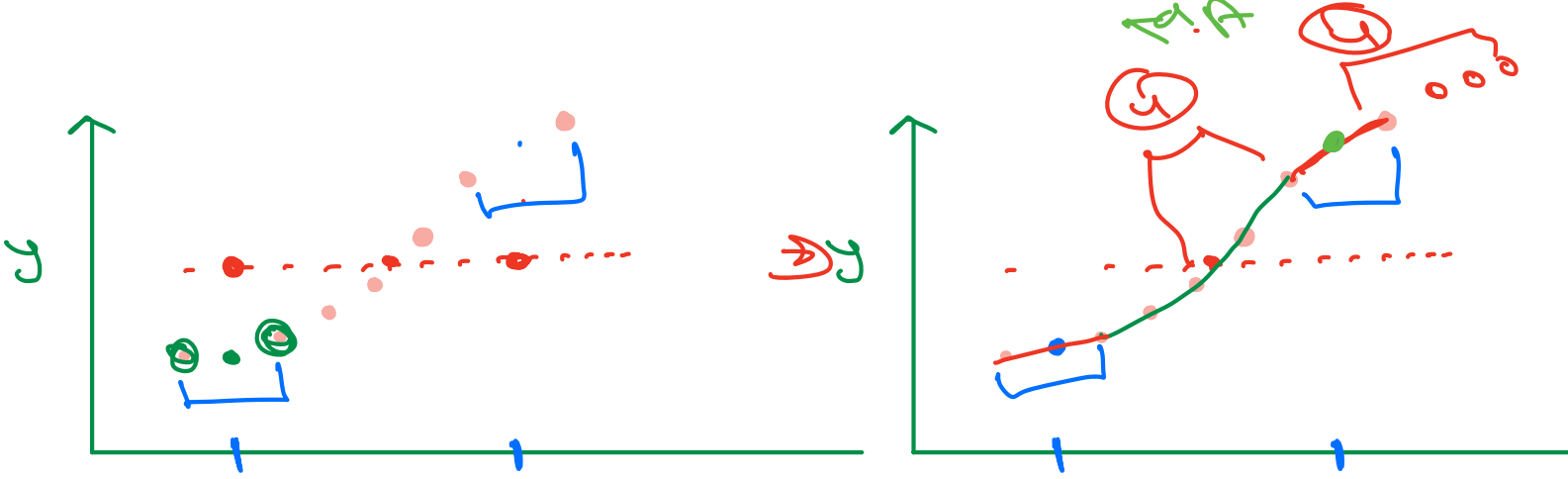


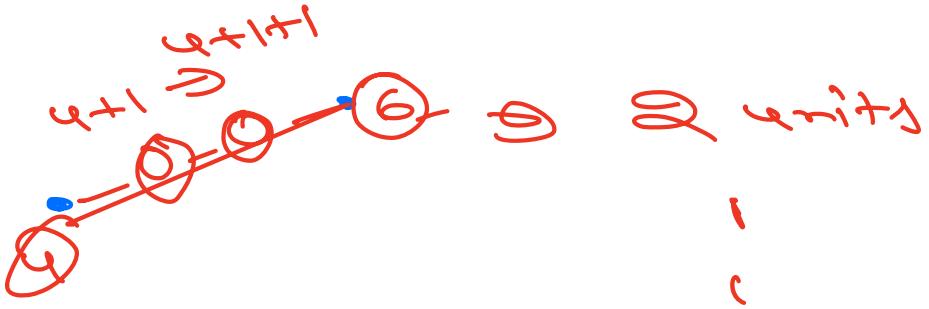
* Interpolation:

Linear Interpolation



$$x_2 = \frac{x_1 + x_3}{2}$$





⇒ 2 units

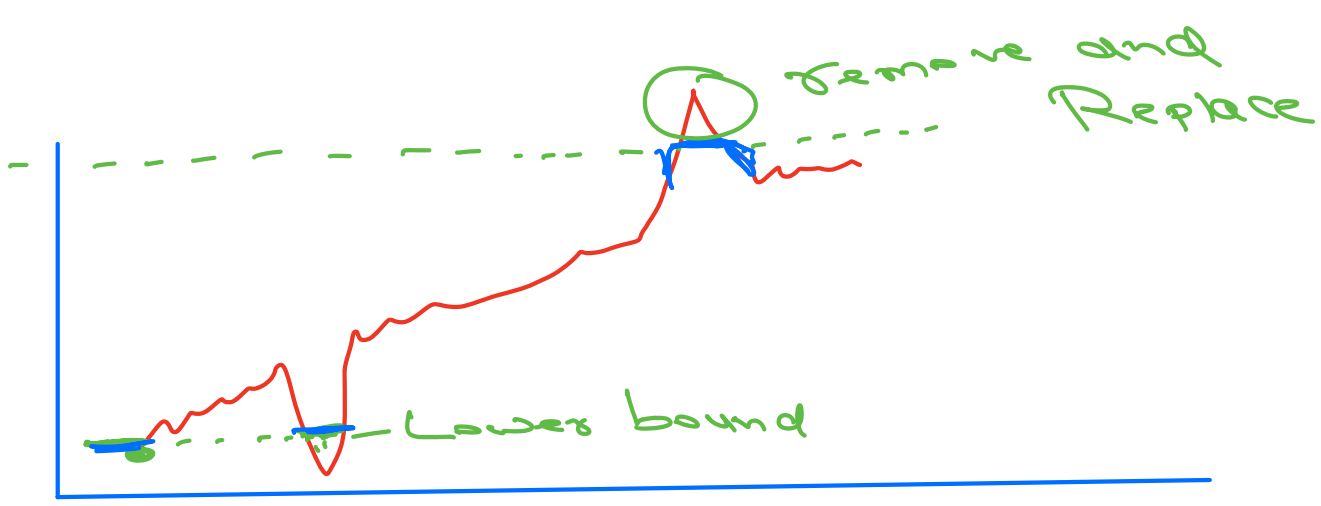
How do we treat Outliers / Anomaly

① Treat them as missing Values and use Interpolation

② Clipping:

Upper bound
- 0.95 percentile
- 0.98 percentile
- 0.995 percentile

Lower bound
- 0.05 percentile
- 0.02 percentile
- 0.0005 percentile

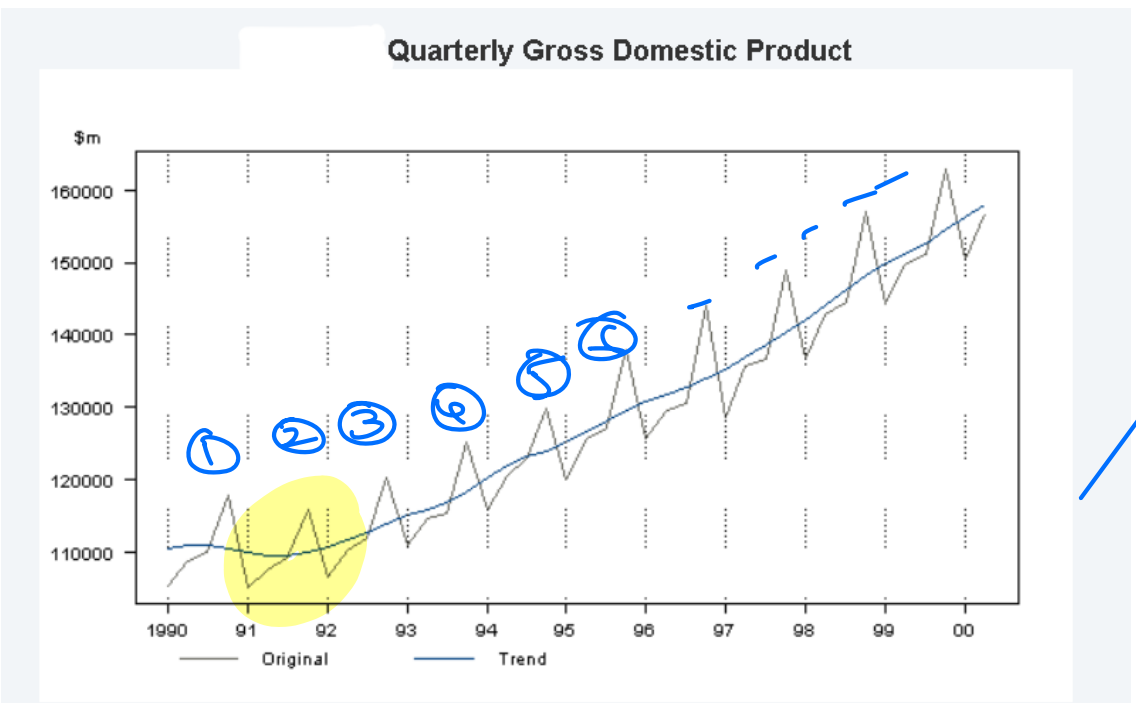


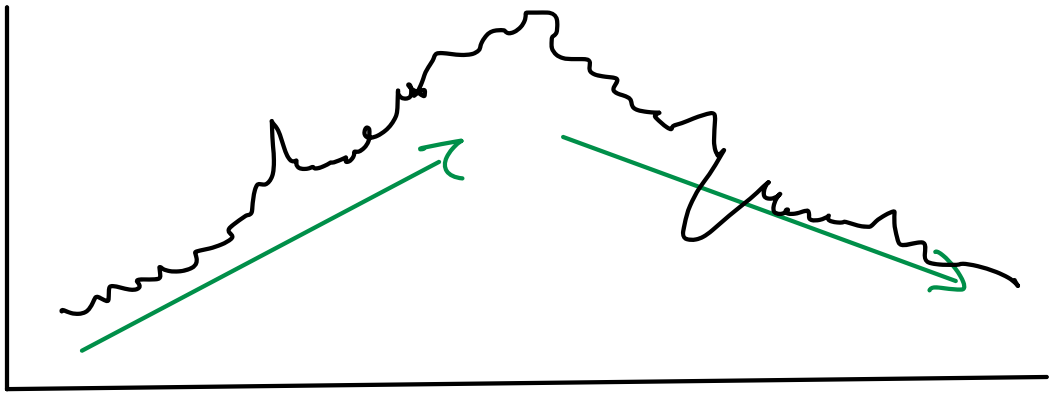
Time-series Decomposition

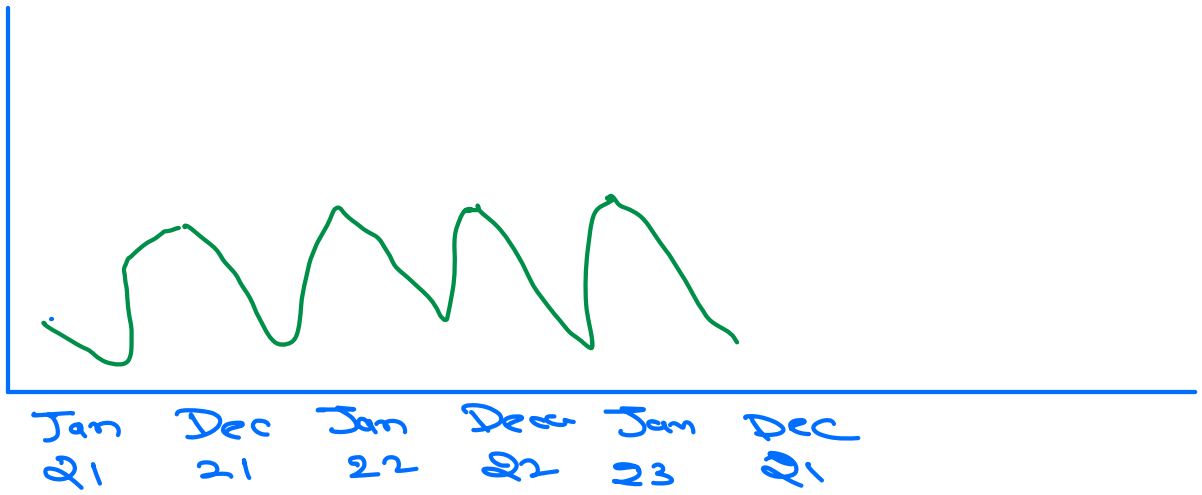
Trend → Is the Target growing or Decreasing

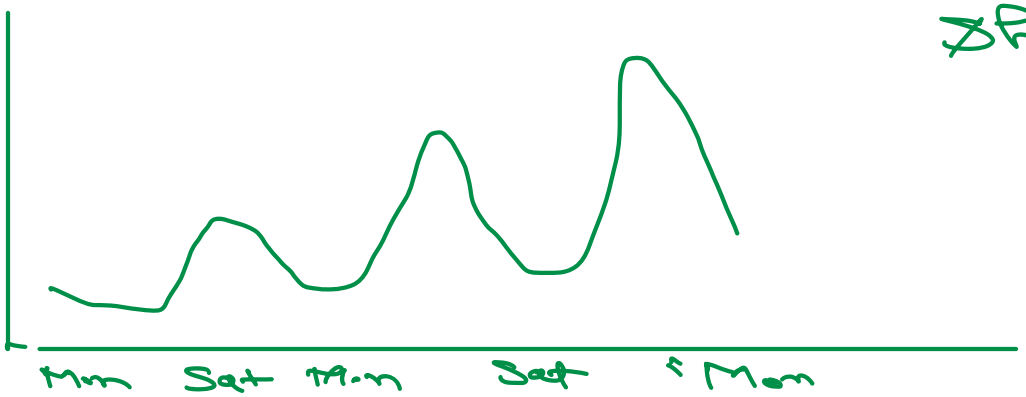
Seasonality → Repeating pattern in Data

Errors



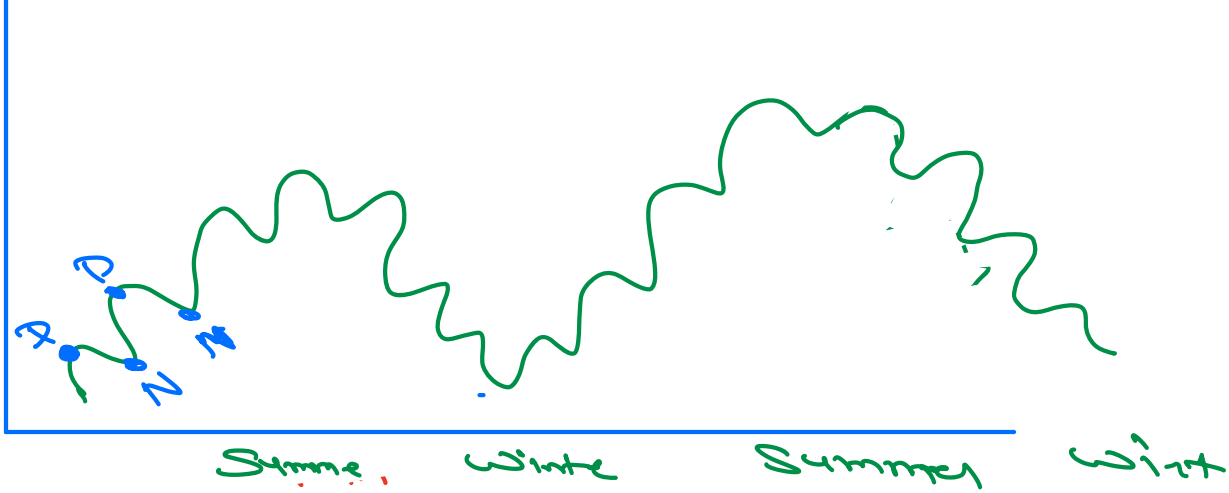






Weekly Season patter

Error Term



Error = T.S - Trend - Seasonality

* Additive Modeling
* We can calculate Trend and Seasonality